

Meeting notes 1:

Supervisor introduced basic concepts of SRCNN, SRGAN, and Real-ESRGAN. Explained the topic and gave directions on brainstorming the topic.

How HQ image degrades to LQ, then CNN 'transfer' back to HQ

Supervisor suggested reading:

[[1501.00092] Image Super-Resolution Using Deep Convolutional Networks](<https://arxiv.org/abs/1501.00092>),

[[1609.04802] Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network](<https://arxiv.org/abs/1609.04802>),

[Real-ESRGAN: Training Real-World Blind Super-Resolution With Pure Synthetic Data](https://openaccess.thecvf.com/content/ICCV2021W/AIM/papers/Wang_Real-ESRGAN_Training_Real-World_Blind_Super-Resolution_With_Pure_Synthetic_Data_ICCVW_2021_paper.pdf)

to brainstorm topic and learn how they work and to be trained.

I was told to play with PyTorch and other related libraries or models to try them out, to have a big picture and better understanding of what I'm going to do.

Meeting notes 2:

I proposed a topic of tennis game video/ image restoration to remove motion blurs and fill in missing gaps due to the low fps of the phone camera. Why is this needed? Even current apps will still blur, balls are too fast, phone camera may not capture the moment the ball touches the ground, hence may not be accurate to judge the ball in or out.

The supervisor rejected my proposed topic, as this is a different level of problem, involves video and filling in the missing piece, which is much harder than the original topic, which will be too difficult for me to do as a capstone project.

At the end, stick to drone image restoration, but need think of what degrade to restore.

Meeting notes 3:

Main purpose is to have a proposal review & advice

had 3 addressed problems: motion blur, low-light, and image size too large

Advice from supervisor: remove image size problem, that is an engineering problem, not a research problem; the addressed problem should be a research problem.

The other addressed problems are fine, since the original real-esrgan focus on natural images, only trains on natural/animated images, not specific to drone image conditions

Strongly suggested work/train model in lab, have GPU, strong enough for the project

The proposal suggested sticking with the handbook guidance.

Change the fonts of the proposal.

For data can find clear, high-quality images, then degrade, then will get image pairs, then train the model with these data.

Meeting notes 4:

The main purpose of this meeting is to explain what I have done so far for the project, consult on the interim report, and discuss the baseline check part of the project. Do I need to output the processed images, review of meeting notes, and report structure, and can I reuse parts from the proposal? Supervisor's feedback is that I should not use Gaussian as a baseline check since it actually makes the image worse, its is the opposite of restoration, and makes the motion blurs worse. Can study LoRA for fine-tuning the model in the next step. And gave suggestions on baseline check on pre-trained Real-ESRGAN.